BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Boolean Conditions in Code

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What are 17 DIV 5 and 17 MOD 5?

2. What happens in the fetch stage?

A) Work out its meaning
B) Get the next instruction
C) Carry out the instruction
D) Save a backup copy

3. "Execution only copies an instruction into memory without performing it." Correct this. Explain why.

4. A CPU compares scores and adds a bonus. Correct both claims. Explain why. (a) The Control Unit stores documents forever. (b) The ALU stores the next instruction's address.

Today's objectives

Use comparisons and Boolean operators in conditions. Use boolean and and in a practical scenario.

Key vocabulary - Read, cover, recall

Boolean
A Boolean value represents true or false.

AND
An AND condition is true only when both component conditions are true.

OR
An OR condition is true if at least one condition is true.

NOT
NOT reverses a Boolean value, changing true to false or false to true.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

What is Boolean?	Boolean - A Boolean value represents true or false.
In this example	A swimming-pool entry policy uses and; An AND condition is true only when both component conditions are true.
Follow the effect	An OR condition is true if at least one condition is true. Explain how this changes the eventual program result.
Check understanding	NOT is used for a specific reason: NOT reverses a Boolean value, changing true to false or false to true.
Boolean is a data type	A Boolean stores true or false. For example, complete=false is a Boolean state; it is different from the string "false". Comparisons such as score>=50 produce Boolean results that can be combined using AND, OR and NOT.

I Do - One worked example

A swimming-pool entry policy is being written. Explain how boolean and and are used to make the program work. [4 marks]

1. We are designing a swimming-pool entry policy. Identify which data and decisions are essential.

2. Boolean [1] - A Boolean value represents true or false. [1].

3. AND [1] - An AND condition is true only when both component conditions are true. [1].

4. Two linked points for each technique give four mark points.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. In a club membership check, explain how boolean helps the program. [2 marks]

2. For a alarm controller, explain why and matters. [2 marks]

3. When designing a timetable access rule, explain how or and not should be used. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the Boolean expression so the result is false when only one check passes.

```
hasPassword = true
hasSecondFactor = false
print(hasPassword ... hasSecondFactor)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State two programming terms or techniques relevant to this lesson. [2 marks]

Q3. A gym opening condition needs to work correctly. Explain how and and or would help. [4 marks]

Q4. A pupil is programming a museum visitor rule. Explain the roles of boolean, or and not in their solution. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A Boolean value represents true or false. ☐ T ☐ F
2. Boolean - A Boolean value represents true or false. ☐ T ☐ F
3. A swimming-pool entry policy uses and; An AND condition is true only when both component conditions are ... ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 DIV gives integer quotient 3; MOD gives remainder 2. ☐

Q2 B - Get the next instruction ☐

Q3 The operation specified by the decoded instruction is carried out. ☐

Q4 (a) The Control Unit decodes instructions and controls CPU operations. (b) The ALU handles arithmetic and logic. Registers store data or addresses briefly. ☐

Code First - Model

```
hasPassword = true
hasSecondFactor = false
print(hasPassword AND hasSecondFactor)
```

Output: false

We Do - Check the model

1. Boolean [1] - A Boolean value represents true or false. [1]. ☐

2. AND [1] - An AND condition is true only when both component conditions are true. [1]. ☐

3. OR [1] - An OR condition is true if at least one condition is true. [1]. NOT [1] - NOT reverses a Boolean value, changing true to false or false to true. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Boolean [1]; OR [1]. ☐

Q3 AND [1] - An AND condition is true only when both component conditions are true. [1]. OR [1] - An OR condition is true if at least one condition is true. [1]. ☐

Q4 Boolean [1] - A Boolean value represents true or false. [1]. OR [1] - An OR condition is true if at least one condition is true. [1]. NOT [1] - NOT reverses a Boolean value, changing true to false or false to true. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: